Business-model diffusion: Phase-0 and first-pass Phase 1

Cross-class theme transit, LDA themes, §5 kill conditions, and a first read of RQ1–RQ4 in four NICE classes (009, 042, 039, 035)

Eric Silver

June 2026 — working note

Abstract

A first, descriptive test of `PROPOSAL_diffusion.md` on a four- class "Uber-for-X" subset — software (009), tech services (042), transport (039), advertising/retail (035), 2.35M granted filings 1990–2024. Three measurements. (1) **Phrase-level transit**: business-model vocabularies originate overwhelmingly in software/tech and arrive in transport and retail years later (cloud +1–4y, AI +2y, as-a-service +3–4y, mobile-app +3y, streaming +5–6y, AR +7y, blockchain +4y). (2) **The filing-level within/cross scalar gap** does *not* detect transit on its own — high-within/low-cross filings are generic vocabulary in an off-class, not coherent imports. (3) A v0 LDA theme model ($T = 50$, fit on a 200k stratified sample, transformed across the full 2.32M-filing pool) **passes both** PROPOSAL §5 kill conditions: the shuffle null is rejected decisively (real mean lag-1 autocorrelation of theme×class prevalence trajectories +0.687 vs. shuffle −0.027, $z = +46$), and the signed-ahead-precedes-behind check is directionally correct (61% of (θ, c) cells negative; mean Spearman $\rho(\Delta\mathrm{KL}, \mathrm{year}) = -0.053$), modest in magnitude. Verdict: the diffusion construct survives the gating tests; the theme is the right unit (not the per-filing scalar); proceed to Phase 1. A first-pass Phase 1 read on the 50-theme model: 46/50 themes cross the 1% floor in at least one class, with 12 reaching all four (RQ1, RQ2); **class-to-class transit is asymmetric and software-led** — 33 of the between-class flows originate in software/tech, only 4 return there. Bass fits ($R^2 \geq 0.7$) on 118 (θ, c) cells give median $p = 0.017$, $q = 0.114$, $q/p \approx 7$, mostly platform-like (RQ3). Provenance among the 50 earliest carriers per theme is 73.7% entrant-side in raw terms (RQ4) but the year-matched population baseline is *higher* at 76.4%, so the mean theme excess is −2.7pp; the raw Schumpeter Mark I read is artifactual — themes inherit the corpus's high overall debut rate without adding to it. A NMF cross-method check ($T = 50$, same vocabulary) replicates the diffusion structure decisively (NMF shuffle null $z = +60.7$) and confirms 14–28% of topics as strongly cross-method-stable (Jaccard $\geq$ 0.3–0.5) — the §2.4 firewall keeps only this subset for headline. The SEC Financial Statement Data Sets panel (2009–2024) now also enables: RQ5 finds lagged class *revenue growth* predicts theme uptake (coef +9.2, $p = 0.048$, $n = 57$), lagged margin does not — direction supports Schmookler/demand-pull. RQ6 finds firms with high filing-year borrowed novelty earn modestly higher contemporaneous gross margin (+1.4pp per σ, $t = 2.1$) but with *lower* R&D intensity — a market-facing innovation signal that *substitutes* for technical invention rather than complementing it.

1 Setup

Four classes chosen to exercise the canonical cross-industry transit: a software goods class (009), a tech-services class (042), and two "physical/commercial" destination classes (transport 039, advertising/retail 035). We condition on **grant** (registration date present; §2.1): 912k/575k/101k/765k granted filings respectively, 2,352,913 pooled overall, with the kill-condition analyses restricted to filing years 1990–2024 (2,317,899 filings).

2 Result 1 — phrase-level theme transit (positive)

For a curated set of multi-token business-model phrases (matched by regex on raw goods/services text, so stopword removal does not mangle "on demand" / "as a service"), we compute per (class, year) prevalence among granted filings and record the first year each class crosses a 1% floor.

Table 1: First year a theme reaches 1% prevalence, by class. "–" = never.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Of the themes that clear the floor in a destination class, almost all originate in software/tech (009/042) and arrive in transport/retail with lags of 1–13 years. The pattern is the proposal's central phenomenon, measured directly. **Honest caveats.** The 1% floor is sensitive to class size (039 is $\sim 8\times$ smaller, so its floor trips on fewer filings); `peer-to-peer` never clears 1% as an exact phrase; and `on-demand` originates in *transport* (2014), a genuine counter-example.

3 Result 2 — the scalar within/cross gap does not detect transit (negative)

For each filing we computed prospective novelty against its own class's past (*within*) and against the pooled-corpus past (*cross*), in a shared 116,783-term universal vocabulary. Within-novelty is typically *below* cross-novelty (own class predicts a filing better than the corpus), so the "high-within/low-cross = borrowed" cell is small (3–5% by class) and its top members — EYEGLASSES DIRECT (009), GAP KIDS (035), EURO-FRAMES (035) — are generic vocabulary that is merely locally uncommon, not coherent business-model imports. The scalar gap conflates "locally rare generic words" with "imported theme," precisely the vocabulary-migration-$\neq$-idea-migration threat (§4.2). It is a useful *filter* (transit candidates should be high-within/low-cross) but not a *detector* on its own.

4 Result 3 — LDA themes pass the §5 kill conditions

Theme model. A 50-topic LDA was fit on a stratified sample of 200,000 granted filings (50k per class, capped at class size; 39c smaller after sampling constraints), with the resulting model used to transform the full 2.32M-filing pool to a per-filing topic distribution. The vocabulary (unigram+bigram, min $df = 200$) had $V = 69{,}361$ terms; LDA fit took 5 min, full-corpus transform 3 min.

Face validity. The top-word lists are recognizable business-model components — e.g., T0 {health, care, wellness, healthcare}; T1 {autonomous, loading, unloading, truck, material handling}; T2 {store, retail, wholesale}; T10 {electronic transmission, computer networks, data, communication}; T30 {online, website, non-downloadable, information}; T49 {vehicles, traffic, bicycle, lights}. We make no claim that 50 unsupervised topics constitute the final theme set — §2.4's cross-method-stability and human-validation firewall is still owed before any individual theme is treated as headline. The kill- condition tests below are the cheap go/no-go check on whether *any* theme extraction can support the program.

Kill condition 1 — shuffle null: pass. For each (theme θ, class c) prevalence trajectory over 1990–2024, we compute the lag-1 temporal autocorrelation. Real themes should have *positive* autocorrelation (smooth ramp-up of adoption); random (class, year) label permutations should average near zero.

Table 2: Lag-1 autocorrelation of (θ, c) prevalence trajectories.

	mean autocorr	cells
Real panel	+0.687	200
Shuffle 1	−0.041	200
Shuffle 2	−0.041	200
Shuffle 3	−0.003	200
Shuffle 4	−0.024	200
Shuffle mean ± sd	-0.027 ± 0.016	—
z = (real − shuffle)/sd		**+46**

The real panel's themes ramp up smoothly within their classes; the shuffled panel is noise around zero. The proposal §5 "date/class shuffle null" is rejected at any plausible significance level. The construct is not an artifact of arbitrary panel slicing.

Kill condition 2 — signed-ahead-precedes-behind: directionally pass, modest. For each (θ, c) cell with ≥ 50 filings carrying θ above $\tau = 0.20$ topic-weight, we compute Spearman ρ between filing-level signed ΔKL (resonance) and filing year. Adoption-curve theory predicts $\rho < 0$: early carriers are signed-*ahead* (resonant; novel language whose class then adopted it), late carriers signed-*behind* (tired; language the class has played out).

(θ, c) cells	197
Mean $\rho(\Delta\text{KL, year})$	−0.053
Share with $\rho < 0$	**61.4**%

A majority of cells move in the predicted direction — the adoption-curve interpretation is not rejected — but the per-cell effect is small. This is consistent with the within-class diagnostics on Class 009 from the construct-validity note: prospective and retrospective KL are 0.97 correlated, so per-filing signed ΔKL is a small residual and a noisy individual adoption-position marker. The aggregate sign comes through; treating signed ΔKL as a clean per-filing curve coordinate is asking more than it can do.

5 First-pass Phase 1: RQ1–RQ4 on the 50-theme model

RQ1 — theme-arrival rate. 46 of the 50 LDA topics cross the 1% prevalence floor in at least one class at least once; 4 remain niche throughout. With the topic count fixed at $T = 50$ the proper Heaps' test is deferred to a variable-K pass, but as a within-fixed-T check: the log-log slope of cumulative-arrived-themes against cumulative filings is $\approx +0.04$ — the arrival is essentially front-loaded, most of the 46 themes are already evident relatively early in the 1990–2024 window.

Table 3: Theme breadth across the four classes.

Classes reached	Themes
1 (niche, class-specific)	13
2	6
3	15
4 (general-purpose)	12

RQ2 — breadth and class flow. 33 themes reach ≥ 2 classes. The class-to-class transit ordering (which class a theme arrives in second, after its origin) is *asymmetric and software-led*:

- software → tech-svcs: 14; software → transport: 11; software → advert/retail: 8 (33 transits originating in software);
- transport → tech-svcs: 7; transport → advert/retail: 6;
- tech-svcs → software: 4 (the only meaningful return flow).

This confirms the phrase-level transit asymmetry of §2 at the theme level: software is a net exporter of business-model vocabulary to the other three classes. With only four nodes a centrality estimate is premature; a 45-class build would let us locate each class in the business-model space.

RQ3 — Bass $p/q/m$ on theme adoption. For each (θ, c) cell where the theme reached at least the floor and was tracked for ≥ 6 years we fit Bass(1969) to the normalized cumulative prevalence and retain fits with $R^2 \geq 0.7$.

Table 4: Bass fits across (θ, c) cells.

Fits retained ($R^2 \geq 0.7$)	118
median p (innovation coef.)	0.017
median q (imitation coef.)	0.114
median q/p	~ 7
"platform-like" ($q/p < 20$)	88
"fad-like" ($q/p \geq 50$)	9

The bulk of the diffusion is platform-like (broad, gradual adoption, moderate word-of-mouth); a small tail of $\sim$ 8% shows fad-shaped curves (very high q/p, sharp imitation pickup). The median $q/p \approx 7$ is in the range typically reported for consumer-product Bass fits, lending external plausibility to the theme×class adoption curves as real adoption events.

RQ4 — entrant vs. incumbent provenance. For each theme we take its 50 earliest carriers (sorted by filing year, conditional on top-topic assignment with weight ≥ 0.20), look up each owner's first granted mark across the four classes, and classify the owner as *entrant* if its first grant is at the carrier-filing year (or one year earlier) versus *incumbent* otherwise.

Themes scored	50
Mean entrant share among 50 earliest carriers	73.7%
Median	73.0%
Themes entrant-dominated ($\geq$ 50% entrants)	48 / 50
Themes incumbent-dominated ($\leq$ 20% entrants)	0 / 50

The raw numbers read as strong Schumpeter Mark I — 48/50 themes entrant-dominated. **But the year-matched population baseline reverses the claim.** Across the 4-class corpus the overall debut rate among granted filings in 1990–2024 is **68–77% per year** (mean 76.4%, weighted by the year distribution of each theme's earliest carriers); the typical filer is making their first grant. Theme-by-theme:

Table 5: RQ4 with population year-matched baseline.

Theme entrant share (mean / median)	73.7% / 73.0%
Year-matched baseline (mean / median)	76.4% / 76.7%
Excess (theme − baseline)	**−2.7**pp / −3.7pp
Themes with positive excess	22 / 50
Themes with excess > 10pp	9 / 50
Themes with negative excess	28 / 50

The raw Schumpeter Mark I read **is artifactual**: themes inherit the corpus's high overall debut rate without adding to it. The distribution of excesses is roughly centered on zero with a small entrant-lean tail (9 themes at +10pp). **What this licenses**: *the typical theme is not disproportionately entrant-born once we account for who files in the relevant years* — a result worth reporting on its own. The 9 themes with real excess merit a closer look (likely the most disruptive, e.g., blockchain/AI-era themes) and are the appropriate locus for any localized Schumpeter I claim. **Methodological note for the program:** this RQ turned on the base rate, which the original §2.6 design did not specify — update the spec to require year-matched baseline for any future provenance-typology result.

6 Cross-method stability via NMF (§2.4 firewall)

A 50-topic NMF was fit on the same 200,000-document stratified sample as the LDA, using the same vocabulary (V = 69,361, min df = 200), and transformed across the full pool. Three comparisons:

Topic-word alignment (Jaccard on top-10 words, greedy match). Median Jaccard = 0.18, mean = 0.24; **28% of NMF topics align with an LDA topic at Jaccard** $\geq$ 0.3**, 14% at** $\geq$ 0.5. The strongly-stable intersection includes textbook business-model components:

Table 6: Top NMF↔LDA aligned topics.

J	pair	top words
1.00	NMF[9]/LDA[16]	energy, electricity, power, renewable
0.82	NMF[14]/LDA[17]	bags, clothing, leather, cases, shoes
0.67	NMF[0]/LDA[2]	services, store, retail, wholesale
0.67	NMF[5]/LDA[7]	music, entertainment, video, audio
0.67	NMF[17]/LDA[28]	systems, control, sensors, devices

The bottom of the alignment is structural: NMF cleanly separates *networks/transmission* (NMF[44]) and *cloud computing* (NMF[2]) where LDA blended cloud computing with lighting/fixtures (LDA[12]); NMF isolates a tight VR/NFT/blockchain topic (NMF[45]) where LDA scattered it. **The §2.4 firewall is therefore not idle: only the $\sim 25\%$ of LDA topics with a Jaccard ≥ 0.3 NMF counterpart should be promoted to headline; the rest are method-artifacts to flag.**

Per-filing top-topic agreement. Adjusted Rand Index = +0.214 on a 500,000-filing sample of paired (LDA, NMF) top-topic labels — modest but well above random (≈ 0). Methods partition the corpus similarly on the stable themes and diverge on the rest, consistent with the Jaccard distribution.

Communication with existing constructs. For nine of ten diff2 seed phrases there is a clear best-NMF-topic:

Table 7: Seed-phrase → NMF topic (best-match by phrase presence in top 50 words).

seed phrase (diff2)	NMF topic (top-50 words contain)
blockchain	NMF[19]: cryptocurrency, blockchain
cloud	NMF[2]: cloud computing
AI / ML	NMF[16]: artificial intelligence, machine learning
SaaS / 'as-a-service'	NMF[12]: software service
mobile-app	NMF[48]: mobile applications
AR / VR	NMF[45]: virtual reality, augmented reality, marketplace
marketplace	NMF[45] (same as AR/VR — a blended topic)
streaming	— no clean NMF topic
IoT	— no clean NMF topic
on-demand	— no clean NMF topic

NMF *independently rediscovers* seven of the ten hand-curated seed phrases as topics. The three missing (streaming, IoT, on-demand) are either too generic to be distinguished from other topics' top words or fall below the cluster threshold at $T = 50$ — worth re-checking at higher T.

Kill condition on NMF: pass, even more decisively than LDA. Re-running the shuffle null on the NMF top-topic panel: real mean lag-1 autocorr = $+\mathbf{0.689}$ (200 cells, essentially identical to LDA's +0.687); shuffle baseline = -0.021 ± 0.012 over three permutations; $z = +60.7$ (LDA was +46). The diffusion structure is *method-invariant* — the trajectories are properties of the corpus, not artifacts of either topic model.

7 RQ5 — margin vs. revenue demand-pull (associational)

Using the SEC FSDS firm-year financial panel (2009–2024; 83,095 firm-year rows, 13,127 unique CIKs) we link USPTO owners to CIKs via the exact-match crosswalk (7,233 matched owners, 4,999 distinct CIKs in the borrowed- novelty panel, 6,628 CIKs with a primary NICE class). For each (class, year) we compute the class-level median gross margin and mean log-revenue growth across firms whose primary NICE class falls in our subset. The outcome — theme inflow — is operationalized two ways: Δ mean share_top and growth in the count of themes above the 1% floor in the class. We regress on lagged margin and lagged revenue growth with class and year fixed effects.

Table 8: Class-level demand-pull regression ($n = 57$, $R^2 = 0.44$).

	coef	t (p)
lagged class margin	−11.34	−1.41 (0.16)
lagged class revenue growth	**+9.22**	**+1.98 (0.048)**

The mean-share-inflow outcome collapses numerically (themes are too sparse for cell-mean changes to read above noise). On the count-of-active- themes outcome, **lagged class revenue growth predicts theme uptake; lagged margin does not.** Direction: *Schmookler / demand-pull* (invention follows market size), not appropriability/rents. **Caveat carried openly:** with 4 NICE classes × ∼14 years = 57 degrees of freedom and class+year FE, this is barely identified and labeled associational. A 45-class build with NICE→NAICS crosswalk would give the proper test; the shock-based causal design (§4.7) is still owed.

8 RQ6 — does borrowed novelty pay? (firm-level outcome bridge)

For the 4,999 CIK-linked firms in the borrowed-novelty panel (21,150 firm–filing-year rows), we aggregated per-filing borrowed-gap to firm-year means and joined to SEC firm-year financials at lag 0, +1, and +2 years. We standardize the firm-year borrowed gap to σ-units and fit firm + year two-way FE with firm-clustered standard errors.

Table 9: Within-firm outcomes per σ of borrowed gap.

	gross margin	rev. growth	R&D intensity
contemp (t)	**+0.014 (t = +2.1)**	+0.011 (0.7)	+0.086 (0.4)
$t + 1$	+0.007 (0.8)	−0.002 (−0.1)	+0.085 (0.3)
$t + 2$	+0.013 (1.6)	−0.008 (−0.6)	−0.142 (−1.0)
cross-section (firm means)	+0.20 (1.3)	−0.003 (−0.2)	**−15.5 (−2.1)**†

†unwinsorized, outlier-prone — treat as suggestive.

Two findings worth keeping: **(a)** contemporaneous gross margin moves up modestly with within-firm borrowed-novelty (+1.4pp per σ, $t = 2.1$), an effect that fades by $t + 1$ — firms that import vocabulary into their class capture some same-year margin lift but it does not persist as a multi-year premium. **(b)** Cross-sectionally, high-borrowed-novelty firms have markedly *lower* R&D intensity — a **coherent picture with the construct-validity finding that ΔKL is a within-firm patent substitute**. Borrowed novelty is the market-facing innovation channel; firms that pull this lever spend less on the technical-invention channel. The two channels appear substitutive within a firm, complementary across the population. This is the localized firm-level innovation signal the original ΔKL critique demanded; the construct earns the bounded innovation reading on this margin-and-substitution shape.

9 What this licenses, and what is next

- The **descriptive diffusion claim is supported** (Result 1), the **construct's gating tests pass** (Result 3), and a first-pass read of RQ1–RQ4 (§5) produces internally consistent diffusion structure: ramped adoption (autocorr +0.69), software-led cross-class flow, Bass-fittable curves with plausible q/p, and a strikingly uniform entrant tilt.

- The **theme is the unit**, not a per-filing scalar (Result 2 vs. 3 together).
- **Phase 1 priorities, ranked by what the data now invites:** RQ2 (breadth and class-to-class theme-flow network — $T = 50$ themes $\times$ 45 classes once we widen the corpus); RQ3 (Bass $p/q/m$ fitted to the transit themes' adoption curves — the trajectories already look Bass-shaped in Result 3); RQ1 (theme-birth Heaps' law); RQ4 (entrant vs. incumbent provenance using the owner panel that is already on disk).
- **Methodological work still owed before headline:** cross-method theme stability (LDA *and* embedding-cluster *and* high-PMI collocations, §2.4); human face-validation on a random theme sample (target ≥ 0.8 rater agreement); enriching the per-cell ahead-precedes-behind test with length-weighting and within-theme-vintage controls (the per-filing signed ΔKL noise issue); a variable-K theme model so RQ1's Heaps reads beyond fixed-T saturation; the §2.6 spec needs to fold in a baseline requirement for any future provenance-typology claim.
- **Now unblocked:** the SEC FSDS panel runs RQ5 (Schmookler-supportive at the 4-class associational level) and RQ6 (modest contemporaneous margin lift, R&D substitute — localized innovation signal on the bounded shape the CV critique was willing to license). Both want a 45-class corpus and a NICE→NAICS crosswalk to read at full power.
- **Open for human input:** `paper/face_validation.md` — a 50-topic rateable form (top words, per-class peak, 8 representative marks, checkboxes). Per §2.4, headline themes need rater agreement ≥ 0.8 on the "coherent" bucket. Combined with the NMF stability filter (Jaccard ≥ 0.3 counterpart, 14/50 strict / 28% loose), the headline theme set is the intersection of cross-method-stable AND human-validated. Currently human-validation pending.

Reproducibility. `scripts/diff1_borrowed_novelty.py`, `diff2_theme_transit.py`, `diff3_lda_themes.py`, `diff4_phase1_rqs.py`, `diff5_rq4_baseline.py`, `diff6_nmf_compare.py`, `diff7_face_validation_doc.py`, `diff8_rq6_borrowed_pays.py`, `diff9_rq5_demand_pull.py`; JSON outputs in `paper/results/{diff1_borrowed,` per-filing panels in `data/processed/{borrowed_novelty,filing_topics,filing_topics_nmf,theme_panel,sec` Corpus: classes 009/042/039/035 from the USPTO TRTYRAP backfile, grant-conditioned, 1990–2024. SEC financial panel: FSDS 2009Q1–2024Q4 (83,095 firm-years; 7,233 USPTO→CIK exact matches).